

DESIGN AND FABRICATION OF AUTOMATED LAWN MOVER

A PROJECT REPORT

Submitted by

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SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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Kuniamuthur, Coimbatore – 641008.

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BONAFIDE CERTIFICATE

Certified that this project report “**AUTOMATED LAWN MOWER**” is the bonafide work of “**AATHIKESAVAN S (727723EUMC001), ABIYUKTH A (727723EUMC004), HARRIN RAJ K (727723EUMC048)**” who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Urban landscapes today face a growing challenge - maintaining lawns while saving time and energy. Homeowners often struggle with uneven grass growth and unpredictable yard conditions that reduce visual appeal and increase effort especially in gardens where the pathway distance is small and limited. AUTO-MOW was developed as a step toward solving this, offering a smarter approach to lawn maintenance and garden care.

The system uses vision-based detection to locate overgrown grass areas and identify zones that require trimming based on height and surface patterns. Alongside, moisture and terrain sensors check into the ground when required to understand softness and surface levels. Once the system detects tall grass, it automatically adjusts the cutting height and speed to be applied, reducing energy usage and unnecessary passes.

To make the solution garden-ready, AUTO-MOW features a compact-drive mechanism that lets the mower move smoothly even between narrow garden paths and flower rows without harming plants. Supported by algorithms and GPS-based navigation, the robot monitors section digitally, ensuring accurate cutting and a sustainable approach to modern lawn care.

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CHAPTER 1

INTRODUCTION

1.1 Background of Automated Lawn Mowing

Lawn care has evolved far beyond traditional methods, with technology now playing a central role in improving efficiency while conserving energy. Automated lawn mowing leverages sensors, data analytics, and automation to make informed decisions, ensuring that operations such as grass cutting, navigation, and obstacle detection are handled efficiently. By monitoring grass and terrain in real time, robotic mowers can respond to conditions faster and optimize cutting patterns, reducing effort and environmental impact. This shift is especially significant for gardens and public lawns, which are maintained on large scales and are highly sensitive to surface variations

1.2 Challenges in Traditional Lawn Maintenance

Traditional lawn mowing often cuts grass uniformly, without considering grass height or terrain variations. This leads to inefficient mowing, higher energy consumption, and uneven lawn appearance. Machines with fixed designs struggle to navigate narrow garden paths, sometimes damaging plants or missing overgrown areas entirely. Human-operated mowing is labor-intensive, inconsistent, and prone to errors, especially in large or uneven lawns, making it difficult to maintain optimal lawn maintenance.

1.3 Importance of Smart Lawn Care Systems

Early and accurate grass monitoring is critical to maintaining healthy lawns. Uneven growth can affect appearance, density, and overall lawn quality. In densely planted gardens, overgrown patches often spread quickly

making timely mowing essential. Smart detection enables targeted cutting, ensuring that only overgrown areas are trimmed, which saves energy, protects surrounding plants, and maintains healthy growth in well-maintained lawn sections.

1.4 Need for Automated Grass Cutting

Grass conditions vary across lawns, affecting growth patterns and surface appearance. Traditional mowing is slow, often involving manual effort and repeated trimming. Automated systems can measure grass height, terrain levels, and moisture in real time, providing insights into lawn conditions and maintenance needs. Integrating this data with navigation systems allows robotic mowers to make informed, timely movements, combining efficient grass cutting with smart lawn management for maximum efficiency.

1.5 Objective of the Automated Lawn Mower System

AUTO-MOW is designed to bridge gaps in conventional lawn maintenance by combining automated grass detection, terrain monitoring, and efficient cutting. Its compact chassis and adaptive wheel mechanism allow it to navigate narrow garden paths without damaging plants. The system integrates machine learning models, vision sensors, and LIDAR mapping to monitor every section digitally, adjust precise cutting patterns, and log maintenance history. The ultimate goal is to maintain lawns, reduce manual effort, and promote Sustainable, smart lawn care practices that are practical for the implementation on the existing real-world landscapes

CHAPTER 2

LITERATURE REVIEW

Bhateja, Neha and Sethi, Nishu and Jain, Shefali and Mishra, Yash, Lawn Mower – An Automated Machine (2020) Robotics is a branch of engineering that combines more than one area of research and is used to design machines that helps us to assist in our day- to-day life. There are various inventions existing that are created using robotics and lawn mower is one such inventions. Lawn Mower or lawn cutter is a mechanical device whose work is to cut grass in our lawns. The automated lawn mower is used to do the same task automatically with help of different sensors and electrical components. This review aims to evaluate the technology that has been not used still to make the system more advanced. This paper presents the currently based system and techniques and retouches the areas that need to be modified for better performance.

This paper presents an automatic lawn mower which is a device or robot that help human to cut grass automatically. Due to rapid development, many conventional lawn mowers have turn into robotic mowers. But they are expensive and have certain demerits. Hence, we designed autonomous lawn mower that fulfills the requirement of a robotic mower, which is economically feasible and environmentally friendly. Sensors are used to provide feedback from outside world. Arduino UNO and Motor Drive Controller microcontrollers controls the entire system. Every action of the Lawn Mower is monitored by the microcontroller with the help of the sensor. Furthermore, Siemens NX as CAD software used to design the structure of the lawn mower. Also, this lawn mower will be self-guided without a need of human directional control due to ultrasonic sensor and the microcontroller that helps in the movement of the machine

Automated Solar Powered Lawn Mower Using Iot,” International Research Journal of Modernization in Engineering Technology and Science, Nov. 2022,

IoT has become a transformative force, impacting various aspects of our daily lives and revolutionizing industries across the globe. This work proposes the design and implementation of an Internet of Things (IoT) based automatic lawn mower system aimed at revolutionizing traditional lawn maintenance practices. The system incorporates cutting-edge technology to enhance the efficiency and convenience of lawn care, catering to the demands of modern urban living.

The core components of the system include a smart lawn mower equipped with sensors, actuators, and connectivity modules, which enable it to operate autonomously. The mower's sensors gather real-time data about the lawn's topography, obstacles, and environmental conditions. This information is processed by an embedded control system, allowing the mower to make intelligent decisions regarding navigation, obstacle avoidance, and mowing patterns. To facilitate user interaction and control, the system integrates with a user-friendly mobile application. This app enables users to remotely monitor and manage the lawn mower, providing features such as scheduling, customization of mowing patterns, and real-time status updates. Additionally, the app may offer weather forecasts and insights into the lawn's health, contributing to a comprehensive lawn care solution.

The work leverages IoT protocols and cloud-based services to enable seamless communication between the lawn mower, user application, and external services proposed a deep learning–based approach for detecting grass height using image analysis techniques. The study introduced a model that combines multiscale feature extraction with contextual attention mechanisms to improve grass identification accuracy. The results showed that the system can efficiently recognize grass conditions under complex lawn environments, supporting automated lawn monitoring and intelligent mowing management.

“Design and Development of Smart Robotic Lawn Mower Using IoT,” Journal of Advanced Engineering and Innovation, Dec. 2023,

IoT has emerged as a powerful technology, influencing multiple domains and transforming automation across various applications. This work presents the design and development of an Internet of Things (IoT) based robotic lawn mower system aimed at improving conventional lawn maintenance methods. The system integrates advanced technologies to increase the efficiency and reliability of lawn care, meeting the requirements of modern environments. The core components of the system include an intelligent lawn mower equipped with sensors, actuators, and communication modules, which allow it to function autonomously. T

he mower's sensors collect real-time data about the lawn surface, obstacles, and surrounding conditions. This data is processed by an embedded control unit, enabling the mower to take intelligent actions related to navigation, obstacle avoidance, and cutting operations. To support user interaction and control, the system connects with an intuitive mobile application.

This application allows users to remotely monitor and control the lawn mower, offering features such as scheduling, customization of mowing paths, and real-time system updates. Furthermore, the application may provide weather information and insights into lawn conditions, contributing to an efficient lawn maintenance system. The work utilizes IoT communication protocols and cloud-based platforms to ensure smooth data exchange between the lawn mower, user interface, and external services presented a deep learning–based method for estimating grass height using image processing techniques. The study developed a model that integrates multiscale feature extraction with contextual attention mechanisms to enhance grass detection accuracy. The results demonstrated that the system can effectively analyze lawn conditions under complex environments, enabling automated lawn monitoring and intelligent mowing operations.

CHAPTER 3

OBJECTIVES

Modern lawn maintenance demands more than just efficiency - it calls for precision, sustainability, and intelligent adaptability. Conventional mowing equipment struggles to navigate narrow garden paths, especially in parks and landscaped areas, often causing unintended plant damage. Simultaneously, the overuse of manual effort and irregular cutting continue to affect lawn quality and reduce visual appeal. AUTO-MOW was conceptualized to overcome these long-standing challenges through balanced integration of robotics, sensing, and artificial intelligence. It aims to deliver a garden-ready solution that not only automates mowing operations but also ensures accuracy, minimal effort, and environmental responsibility

3.1. Specific Objectives

1. **To engineer a compact chassis system** equipped with an adaptive mobility mechanism that allows smooth traversal through narrow garden paths without damaging surrounding plants.
2. **To reduce uneven grass cutting** by enabling location-specific mowing only where overgrown grass is detected, ensuring optimal cutting efficiency.
3. **To prevent incorrect mowing patterns** by using AI-driven grass detection models that match the detected grass height with the appropriate cutting level and movement speed.
4. **To enable real-time lawn condition** evaluation using retractable height and humidity sensors that touch the grass surface when required, avoiding unnecessary contact

5. **To incorporate vision-based imaging** for grass detection based on distinct height variations and color patterns, ensuring precise identification even in early growth stages.
6. **To build a digital map of the lawn area** using LiDAR scanning for spatial mapping, allowing users to visualize trimmed zones and monitor mowing history
7. **To promote sustainability** by minimizing energy consumption, preserving surrounding plants, and ensuring healthy lawn growth over multiple maintenance cycles.
8. **To contribute to smart landscaping** research by providing a scalable model that integrates advanced sensing, mapping, and automation technologies into a single affordable lawn system
9. **To enhance efficiency and decision-making** through machine learning algorithms that interpret sensor data and generate optimized mowing paths for future lawn maintenance.

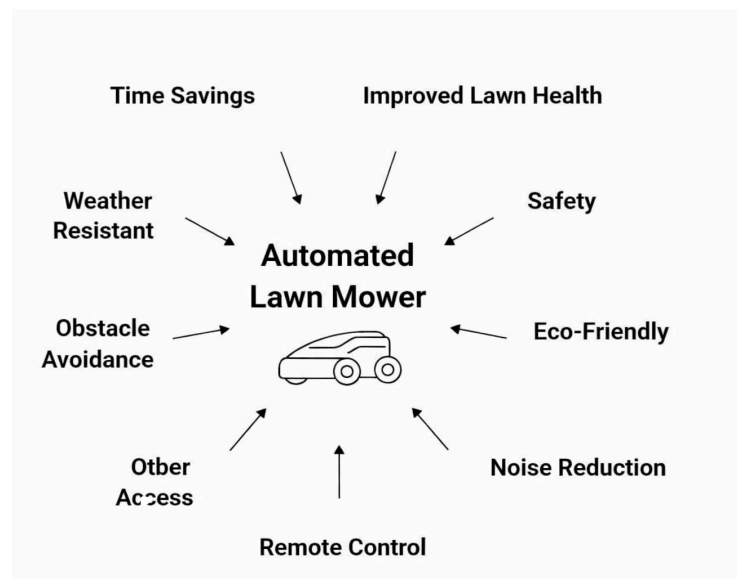


Fig. 3.1.1. Objectives

CHAPTER 4

METHODOLOGY

4.1 System Architecture Overview

The AUTO-MOW architecture follows a hierarchical control model. The Raspberry Pi functions as the master controller, handling high-level navigation planning, image processing, and cloud synchronization through AWS IoT Core. The Arduino Mega manages low-level actuation, blade speed regulation, and motor control tasks. Communication between both units is established via a UART interface, while LoRa R-02 modules enable long-range wireless data exchange across the lawn. Lawn data, image logs, and sensor readings are uploaded to the AWS Cloud for storage, analytics, and mobile accessibility through a React Native-based user interface.

4.2 Mechanical Design of Lawn Mower

The mechanical chassis is compact, lightweight (≈ 12 kg frame) and designed for narrow-path navigation in gardens and lawn areas. It employs an adjustable cutting mechanism that raises or lowers the mowing blade between 20–45 mm according to grass height, eliminating unwanted contact with soil. The base uses 150–200 mm diameter rubberized wheels for traction on uneven or damp ground surfaces. The cutting mechanism is powered by a DC motor (12 V, high speed) driven by the Arduino's motor driver module. The chassis frame is constructed using mild steel square tubing (20×20 mm) to provide a strong.

yet lightweight structure with high corrosion resistance, suitable for humid and outdoor conditions. The design includes a central mounting rail that supports interchangeable sensor and cutting modules, enabling modular maintenance and scalability. A dual-wheel differential drive system allows precise turns even within confined spaces, while a caster-assisted rear wheel stabilizes motion across uneven lawn terrain. The overall weight distribution has been optimized to maintain a low center of gravity, improving balance during operation on sloped lawns. To ensure durability, the frame is coated with anti-rust epoxy paint, and all fasteners are stainless.

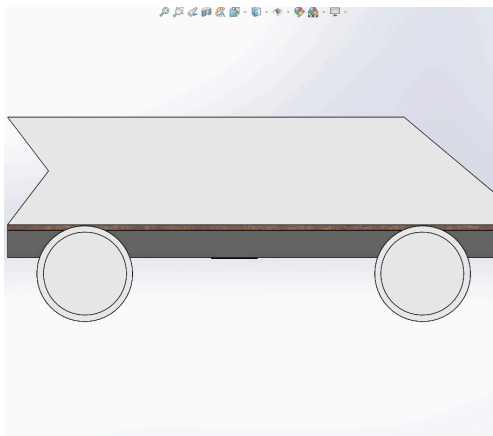


Fig 4.2.1. Right View

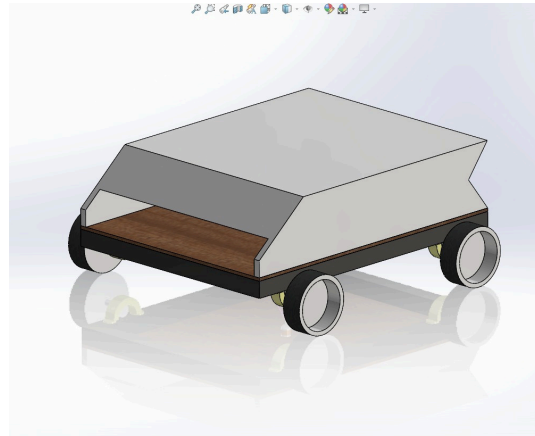


Fig 4.2.2. Side View

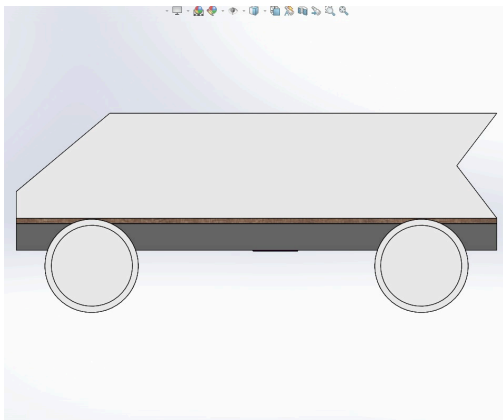


Fig 4.2.3. Left View

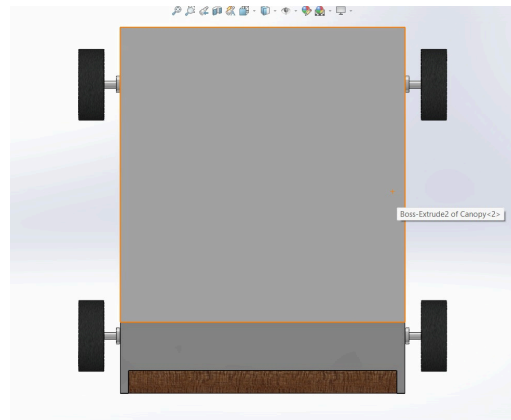


Fig 4.2.4. Top View

4.3 Sensor Integration

The sensing module integrates:

Ultrasonic and ground sensors that periodically scan the surface using a servo-controlled mount to measure grass height, terrain distance, and surface obstacles.

RS-485 based grass height and moisture sensors that periodically scan into grass surface using a servo-controlled probe to measure height, density, thickness, and moisture content.

pH sensor for lawn surface acidity / alkalinity monitoring.

LiDAR unit (360° range ≤ 12 m) for terrain mapping and digital twin generation of the field. All sensor data streams are timestamped and transmitted to the Raspberry Pi for pre-processing before cloud synchronization.

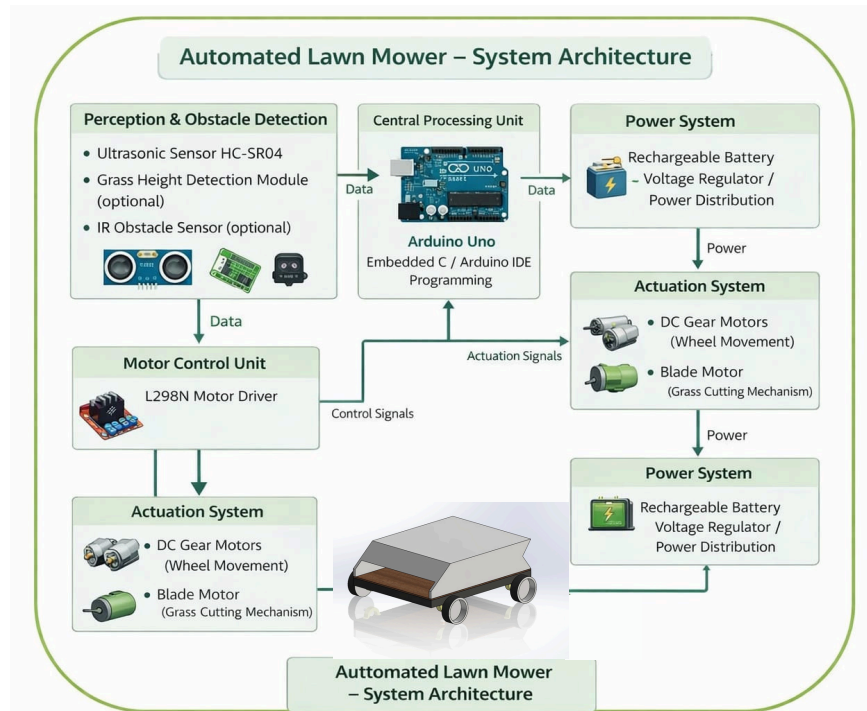


Fig. 4.3.1 Sensor Integration

4.4 Sensor Integration (Obstacle Detection)

A CNN-based classification model was developed using over 6,000 labeled grass images of lawns captured in garden and park conditions. Data were augmented with variations in lighting, angle, and scale. Feature extraction utilized visual texture combinations to detect height differences corresponding to overgrown grass. The trained model achieved 92.8 % accuracy in grass classification on validation data. The model runs locally on the Raspberry Pi using TensorFlow Lite, allowing offline inference and immediate mowing action without network dependency

4.5 Microcontroller and Control Logic

The master software stack is written in Python 3.11 and manages sensor polling, navigation inference, and motion control commands. The Arduino Mega executes low-level routines in C++, controlling motors, blade drivers, and wheel mechanisms. Data is sent to AWS S3 and DynamoDB for storage; analytics dashboards are hosted through AWS QuickSight. The React Native mobile app provides users with live visuals, lawn status reports, and mowing activity logs. Control logic follows a three-tier loop: (i) data acquisition → (ii) analysis → (iii) motion execution with feedback validation. To maintain real-time responsiveness, the system uses a multi-threaded architecture, allowing simultaneous data collection and motor control without latency. A priority-based interrupt system ensures that safety-critical commands, such as emergency stop or blade halt, override routine operations. Additionally, all system events and sensor anomalies are logged locally and in the cloud for predictive maintenance and performance analytics, improving long-term reliability and traceability.

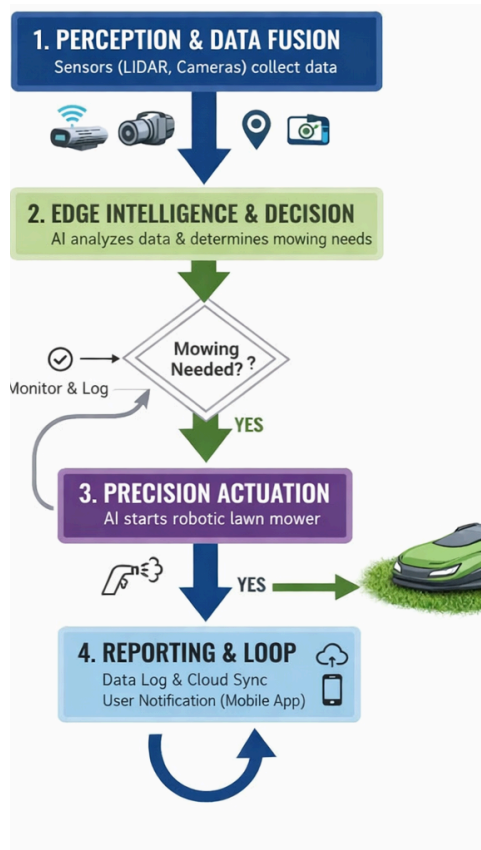


Fig 4.5.1 Operational Flow

4.6 Power Supply and Battery System

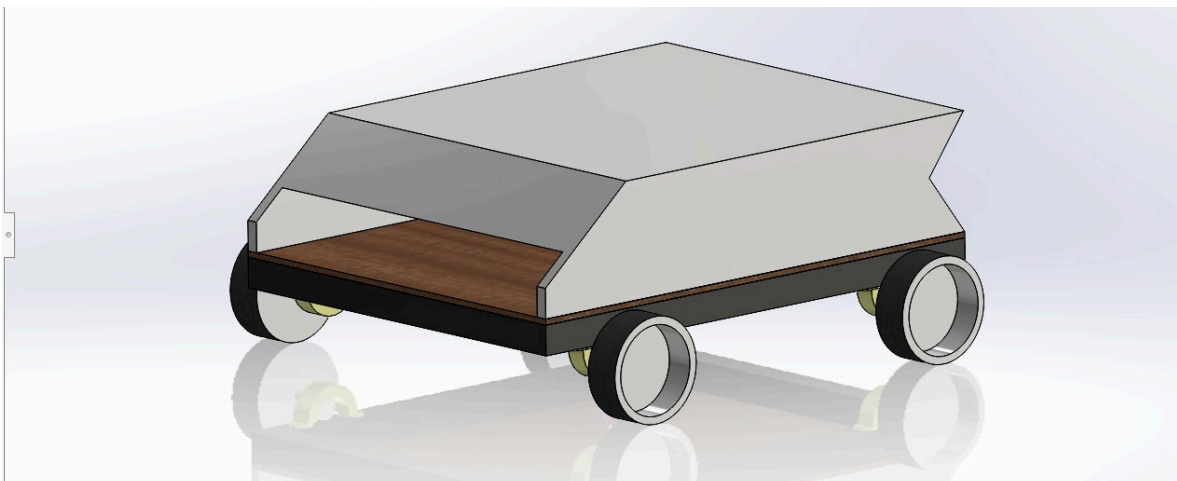
Once overgrown grass is detected, the Raspberry Pi calculates mowing coordinates and commands the blade motor controller via Arduino. The precision cutting system operates through a 12 V high-speed blade motor connected to a PWM-regulated driver, allowing speed modulation (2000–3500 RPM) based on grass density. The system supports multiple cutting heights (up to 3 levels) and automatically adjusts to the recommended level. Mowing logs are tagged with mapped coordinates for traceability in the field map.

4.7 Navigation and Movement Mechanism

Power is supplied by a 12 V 15 Ah Li-ion battery pack supporting ≈ 6 hours of field operation. Voltage regulation (5 V/3.3 V rails) is managed through buck converters for logic components. LoRa provides a communication range up to 2.5 km in open fields, ensuring seamless coordination between the robot and the base station. Fail-safe communication fallback uses onboard Wi-Fi (IEEE 802.11 b/g/n) when close to the farmhouse network.

4.8 Safety Features and Protection System

Each mowing operation generates a synchronized safety response containing sensor readings, obstacle alerts, and movement coordinates. These are locally buffered in SQLite and periodically monitored by the control system. The safety module performs obstacle detection, collision avoidance, and emergency response for operational protection. Users can review obstacle alerts, system status charts, and safety reports directly on the mobile dashboard, allowing long-term monitoring of lawn safety and system reliability.



CHAPTER 5

PROJECT WEBSITE AND DIGITAL OUTREACH

5.1 Introduction

In modern engineering projects, presenting the work through digital platforms has become essential for wider accessibility and effective communication. To support the AUTO-MOW intelligent lawn mowing system, a dedicated project website was developed. The website serves as a digital platform to present the concept, technical details, working mechanism, and significance of the project in an organized and easily understandable format.

The website was created to ensure that the project is not limited to documentation alone but is also accessible to researchers, students, professionals, and potential collaborators. By hosting the project information online, the team aims to make the innovation visible to a broader audience and demonstrate its practical relevance in the field of precision agriculture.

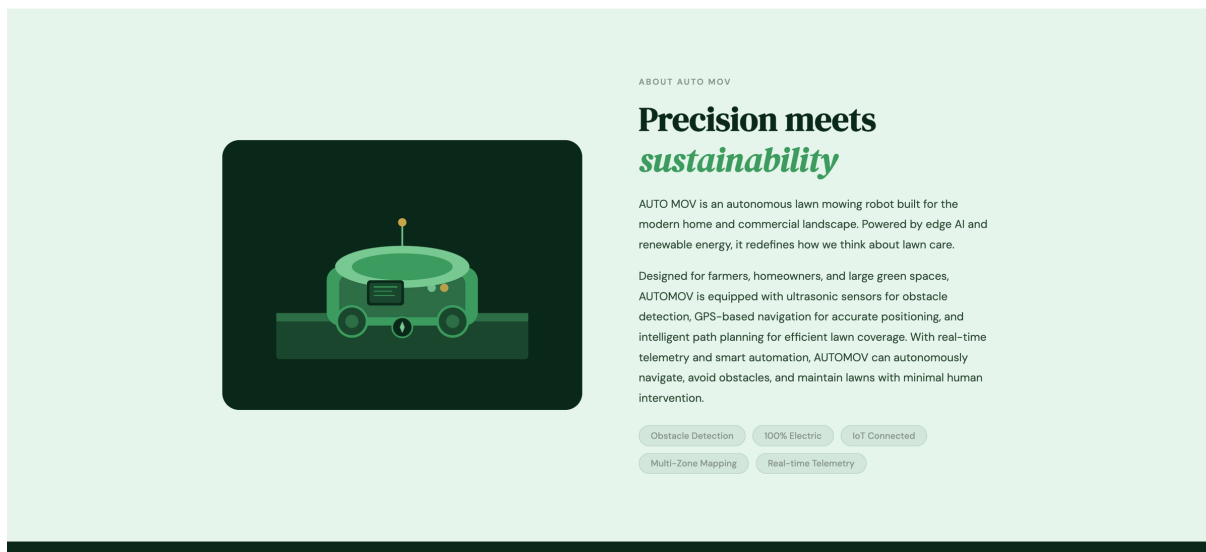


Fig 5.1.1 Home Page

5.2 Website Development

The project website was designed and developed by the team to present the AUTO-MOW system in a visually structured and informative manner. The platform includes sections that explain the problem addressed by the project, the motivation behind the development of the system, and the technological approach used to build the intelligent lawn mowing solution. Special attention was given to the clarity of information and the presentation of visuals so that visitors can easily understand the working of the system. The design approach focused on simplicity, readability, and accessibility while ensuring that the technical aspects of the project are communicated effectively. The website can be accessed at: <http://127.0.0.1:5500/AUTOMOV.html#about>

5.3 Website Content Structure

The website contains multiple sections that collectively describe the project in detail. These include:

- Project Overview – Provides a brief introduction to the AUTO-MOW system and the motivation behind its development.
- Problem Statement – Explains the challenges associated with traditional lawn mowing and the need for intelligent automation in lawn maintenance.
- Solution Description – Describes the proposed robotic system and how it integrates sensing technologies, machine learning, and automated cutting.

System Features – Highlights the important components such as multispectral imaging, LiDAR mapping, and the adjustable chassis mechanism.

Visual Demonstrations – Includes images, diagrams, and design views of the prototype to provide a better understanding of the system information easily and gain a comprehensive understanding of the project.

This structured presentation helps users navigate through

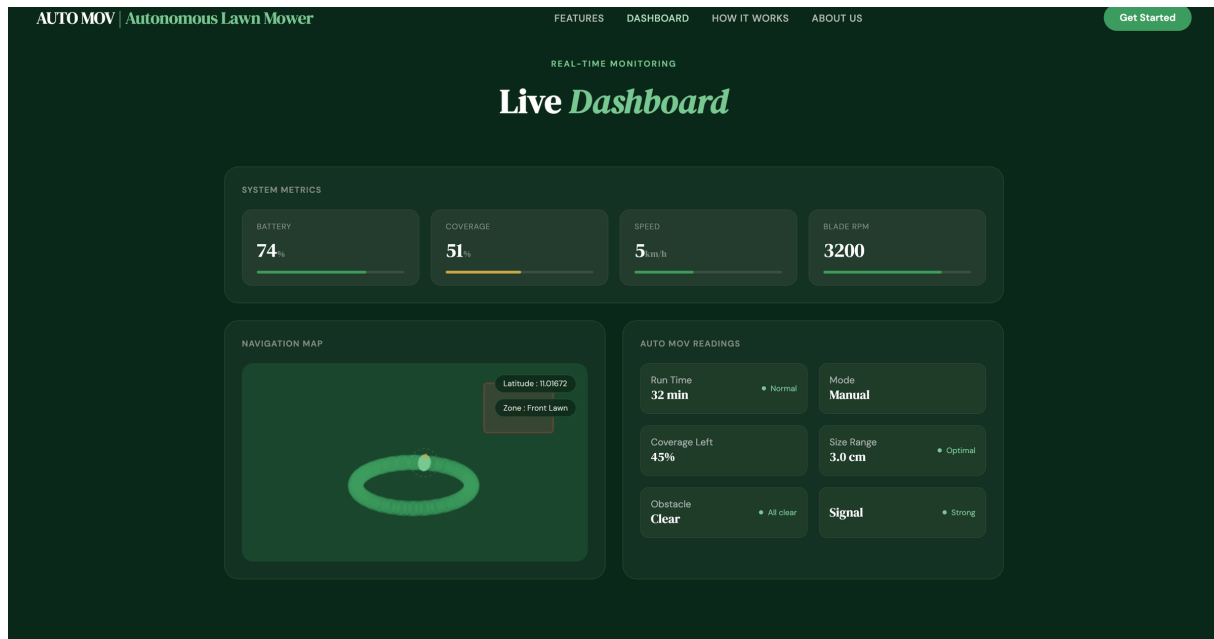


Fig 5.3.1 Dashboard

5.4 Importance of the Website for the Project

The project website plays a significant role in extending the impact of the work beyond the academic environment. It provides a platform where the innovation can be viewed, evaluated, and understood by a wider community. This is particularly useful for projects related to

smart lawn care, where real-world adoption and collaboration are important. Through the website, the project team can demonstrate the practical relevance of the AUTO-MOW system and showcase how modern technologies such as artificial intelligence, sensors, and automation can be integrated to solve lawn maintenance challenges. The platform also supports knowledge sharing and encourages discussions about future improvements and applications.

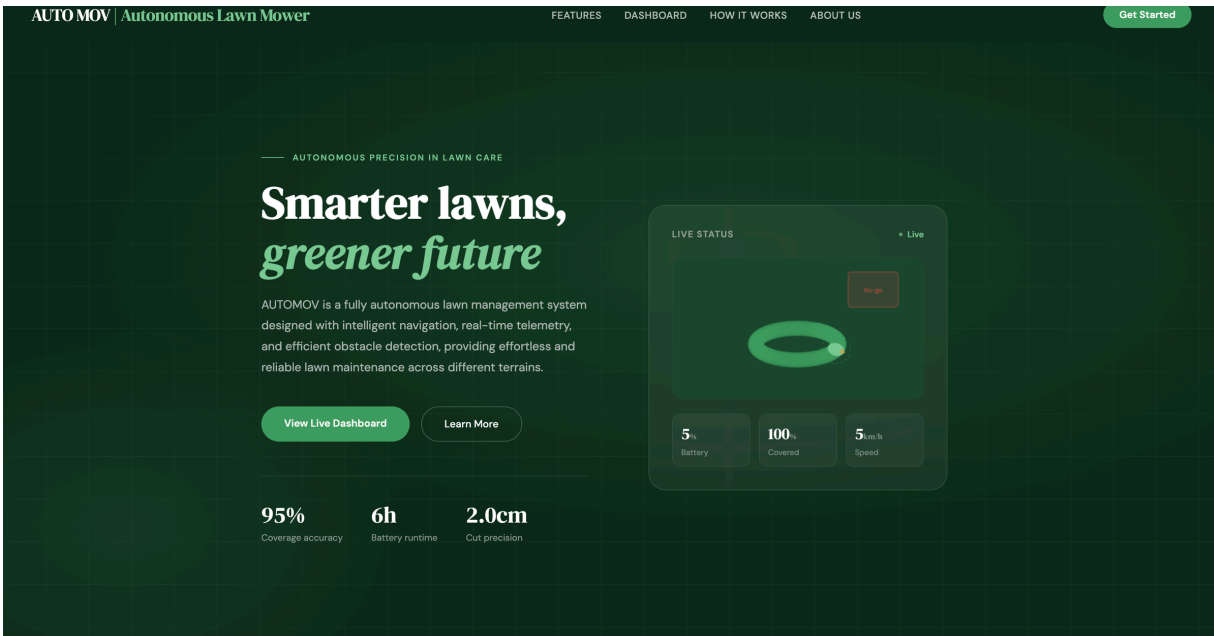


Fig 5.4.1 Home Visit

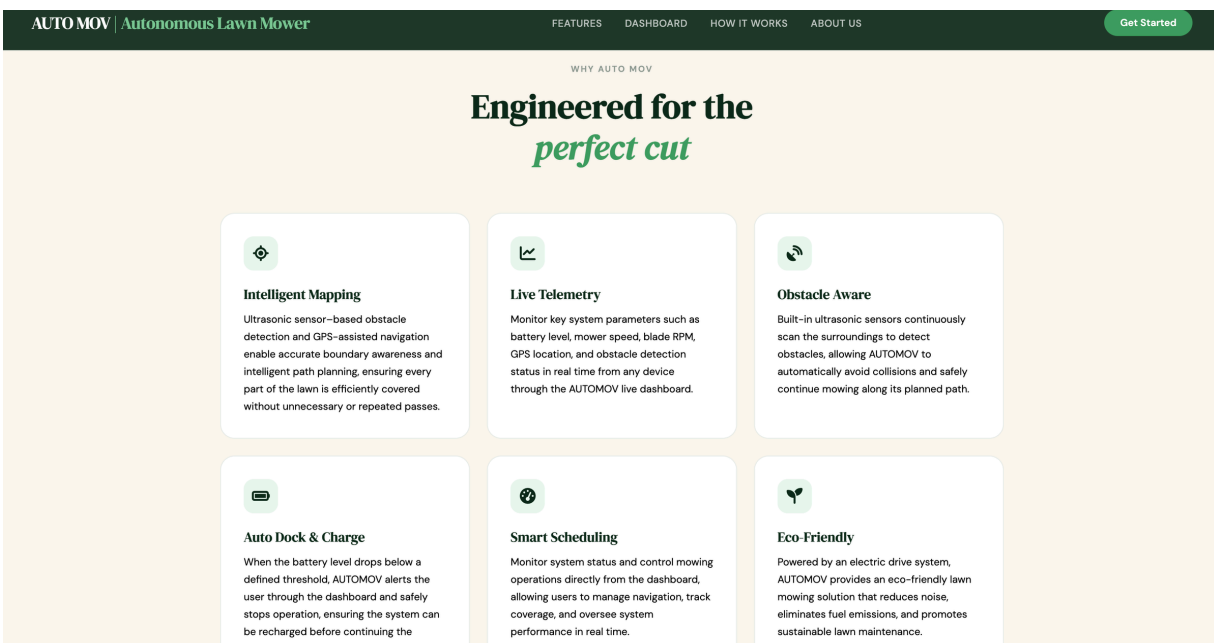


Fig 5.4.2 Feature

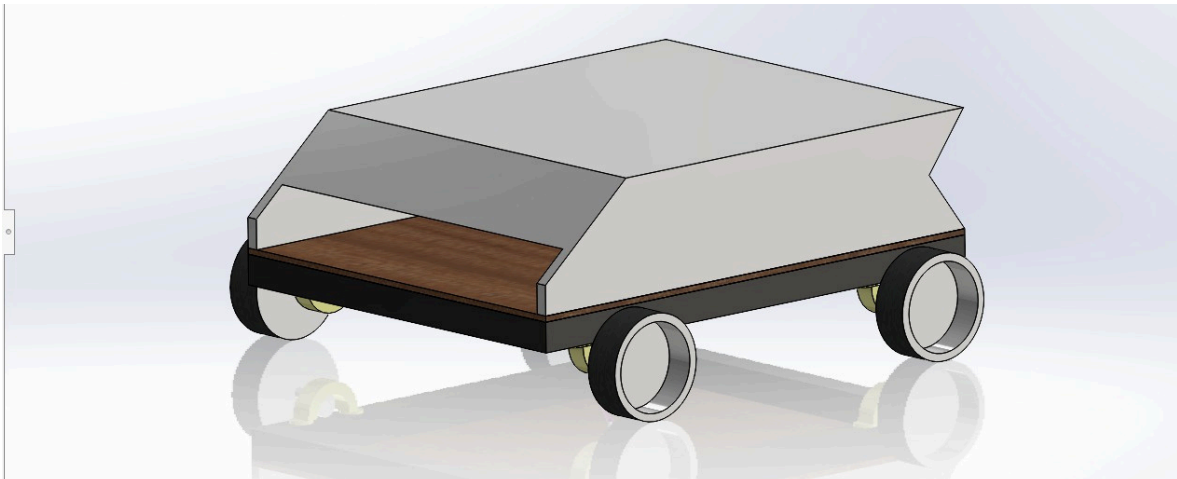


Fig 5.4.3 CAD Model


AUTOMOV

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Send us a Message

Full Name *

Email Address *

Company/Farm Name

Message *



Contact Information

Questions about our system? Need technical support? Contact us through any of the channels below.


Email

contact@automov.in
 support@automov.in
 Drop us a line anytime


Phone

+91 79044 63018
 +91 95858 43031
 Call us during work hours


Address

SKCET
 Coimbatore – 641008
 Tamil Nadu, India

Fig 5.4.4. Contact Page

CHAPTER 6

COST ESTIMATION & GANTT CHART

6.1 Cost Estimation

Table 6.1.1 Cost Estimation (approx.)

S. No	Component	Specification / Description	Quantity	Approx. Cost per Unit (₹)	Total Cost (₹)
1	Arduino Uno	Main microcontroller for control and communication	1	500	500
2	Motor Driver (L298N)	For controlling wheel movement	1	300	300
3	DC Geared Motor	For chassis movement	2	600	1,200
4	Wheels	Rubber wheels with hub	4	250	1000
5	Chassis Frame	Mild Steel	1	1,000	1,000
6	Multispectral Sensor	for lawn height	1	5,000	5,000

CHAPTER 7

CONCLUSION

Modern lawn maintenance requires efficient and sustainable solutions to manage grass growth, reduce manual effort, and improve lawn quality. The AUTO-MOW intelligent lawn mowing system was developed to address these challenges by integrating automation, sensing technologies, and data-driven decision making into a single lawn care platform. The system combines vision-based imaging, machine learning, terrain sensors, and LiDAR-based mapping to monitor lawn conditions and detect uneven grass growth at an early stage. The project demonstrates the design of a compact robotic system capable of navigating narrow garden paths while performing precise grass cutting. The adaptive drive mechanism enables the system to operate without damaging surrounding plants, while the integrated sensing modules collect real-time data on grass height and surface conditions. Based on this information, the system adjusts and performs the appropriate cutting action in precise levels, reducing energy wastage and improving mowing efficiency. By combining grass detection, terrain monitoring, and automated mowing within a single platform, AUTO-MOW supports the goals of smart landscaping and sustainable lawn care. The project highlights the potential of integrating robotics, artificial intelligence, and sensor technologies to create practical solutions that enhance lawn maintenance, optimize resource usage, and contribute to the future of intelligent outdoor automation.



SUSTAINABLE DEVELOPMENT GOALS MAPPING

The Sustainable Development Goals (SDGs) are a set of 17 global objectives established by the United Nations in 2015 to create a more sustainable and better future for all by 2030. Adopted by 195 countries, these goals provide a universal framework for addressing global challenges such as sustainability, innovation, and environmental protection.

The AUTOMOV autonomous lawn mower project is aligned with these goals, particularly focusing on sustainable urban development, technological innovation, and eco-friendly solutions. By integrating automation and energy-efficient systems, the project contributes to building smarter and more sustainable environments.

Questions	Answers
Which SDGs does the project directly address?	SDG 9 - Industry, Innovation and Infrastructure SDG 11 - Sustainable Cities and Communities SDG 12 - Responsible Consumption & Production
What strategies or actions are being implemented to achieve these goals?	Development of an autonomous navigation system using sensors (ultrasonic, GPS, etc.) Use of smart algorithms for obstacle detection and path optimization
How is progress measured and reported in relation to the SDGs?	Area covered per charge, Energy consumption efficiency, Obstacle detection accuracy, Autonomous navigation success rate, Maintenance reduction
How were these goals identified as relevant to the project's objectives?	The project focuses on automation, sustainability, and efficiency, which directly align with SDGs related to: SDG 9,11,12
Are there any partnerships or collaborations in place to enhance this impact?	Academic collaboration, Industry partnerships, Startup ecosystem, Government / Smart City initiatives

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